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Ultra-Wideband 4-Bit Distributed Phase Shifters Using Lattice Network at *K/Ka*- and *E/W*-Band

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ABSTRACT In this article, we introduce an ultra-wideband 4-bit distributed phase shifter using a lattice network. To achieve wider bandwidth, the proposed phase shifter employed an all-pass lattice network instead of the traditional low-pass ladder network. Seven cascaded 22.5° lattice phase shifters and one switched line 180° phase shifter were used to achieve 360° phase shift range. Based on our theoretical analysis, we designed the lattice network as a constant-phase shifter rather than a delay line. Implementations in the *K/Ka*- and *E/W*-bands validate the suitability of the lattice network for constant-phase shifting. Fabricated using 28-nm bulk CMOS technology, the *K/Ka*-band phase shifter had a size of 0.45 mm^2 excluding pads. Within the frequency range of 20.5–35.5 GHz, the root-mean-square (RMS) phase error ranged from 1.6 to 5°, the RMS gain error ranged from 0.3 to 0.6 dB, and the return loss remained above 10 dB. At 28 GHz, the insertion loss was 11.6±0.8 dB without dc power consumption. Fabricated using 28-nm FD-SOI technology, the *E/W*-band phase shifter had a size of 0.3 mm^2 excluding pads. Within the frequency range of 63.5–100.5 GHz, the RMS phase error ranged from 2.4 to 4.6°, the RMS gain error ranged from 0.44 to 1 dB, and the return loss remained above 10 dB. At 82 GHz, the insertion loss was 11.9±1.1 dB without dc power consumption. The proposed phase shifter demonstrated exceptional performance for multistandard operation, achieving low RMS phase and gain errors across a wide fractional bandwidth of 53.6% and 45.1%, respectively.

INDEX TERMS CMOS integrated circuits, lattice network, millimeter-wave integrated circuits, passive circuits, phase shifters, wideband.

I. INTRODUCTION

THE RAPID expansion of data communications and multimedia applications is fueling the demand for high-speed communication technologies in 5G millimeter wave (mmWave) [1], [2]. Phased array systems enable beam steering and reduce severe path loss, enhancing their suitability for 5G mm-wave communications with short propagation distances [3]. However, the frequency spectrum allocated for 5G mmWave communication varies across countries, necessitating multistandard phased array systems for global operations and cost-effective deployment [4]. A multistandard phased array system requires uniform beamforming performance across narrow multiband signals. To meet this requirement, multistandard phased array systems must implement a wideband constant-phase shifter that provides equal phase shifts between multiband signals. Unlike true-time delay (TTD), in which phase shift varies with frequency, constant-phase shifters are better suited for beamforming narrow multiband signals. Moreover, achieving precise beamforming in multistandard phased array systems requires low root-mean-square (RMS) phase/gain error and high return loss over a wide bandwidth. The phase shifter, which is one of the key components in a phased array system, can be implemented as either an active or passive circuit. Active phase shifters, such as active vector modulated phase shifters (A-VMPSs), offer high gain and compact size [5], [6], [7], [8], [9], [10]. A-VMPSs control the in-phase (I) and quadrature (Q) path signals through the variable gain amplifiers (VGAs) to achieve the desired phase shift. However, the use of VGA introduces limitations to the linearity of the A-VMPS and the overall linear output power suffers from the inherent 3 dB losses within the combiner. In the millimeter-wave frequency band, X-type attenuator (X-ATT)-based passive vector modulated phase shifters (P-VMPSs) have recently been widely studied [11], [12], [13], [14], [15]. P-VMPS has the advantages of low RMS phase/gain error, wide bandwidth, zero power consumption, and high linearity. However, achieving phase shifting through the combination of attenuated I/Q signals results in high insertion loss and bandwidth limitations due to the I/Q generator. Switched-type phase shifters (STPSs) [16], [17], [18], [19], [20], [21], [22] are also one of the main passive phase shifter topologies. STPSs enable phase switching by switching between artificial transmission lines with distinct electrical lengths. However, the low-pass or high-pass characteristics of artificial transmission lines, based on T or π networks, lead to reduced bandwidth in STPSs. The reflective-type phase shifters (RTPSs) with varactors offer the advantages of low insertion loss and continuous phase shift [23], [24], [25]. RTPS achieves phase shifting through a variable capacitor in the reflective load. However, the resonant structure of a reflective load introduces drawbacks, such as large phase errors, large insertion loss variation, and relatively narrow bandwidth.

In Fig. 1(a), the configuration of a differential distributed phase shifter is shown. This configuration allows the creation of a 360° N-bit phase shifter by cascading 2^N-1 stages of $\Delta\theta$ unit phase shifters. The advantage of distributed phase shifters lies in their ability to minimize frequency-dependent device parasitics between $\Delta\theta$ unit phase shifters, resulting in a broad impedance-matching bandwidth [26], [27]. This broad impedance-matching property has enabled the use of distributed networks for a variety of applications, including phase shifters and TTD [28], [29], [30], [31]. However, distributed networks are typically designed as ladder networks. This causes the unit networks to behave as delay lines rather than constant-phase shifters, resulting in narrow phase shift bandwidths. Therefore, the bandwidth of distributed phase shifters can be increased by using unit phase shifters that operate as constant-phase shifters rather than delay lines.

In this article, we propose a unit phase shifter based on an all-pass lattice network as shown in Fig. 1(b). Previously studied all-pass lattice networks have been studied as delay lines that offer wide impedance-matching bandwidth. However, as shown in Fig. 1(c), the lattice network can also be operated as a constant-phase shifter with opposite phase characteristics with delay lines. To enhance the bandwidth of the distributed phase shifter, we replaced the ladder network with an all-pass lattice network. Importantly, the wideband characteristic inherent in the all-pass lattice phase shifter is maintained when it is cascaded into a distributed phase shifter. Previously, we proposed a wideband differential phase shifter for the *K/Ka*-band using a lattice network [32]. We extended our previous work to implement

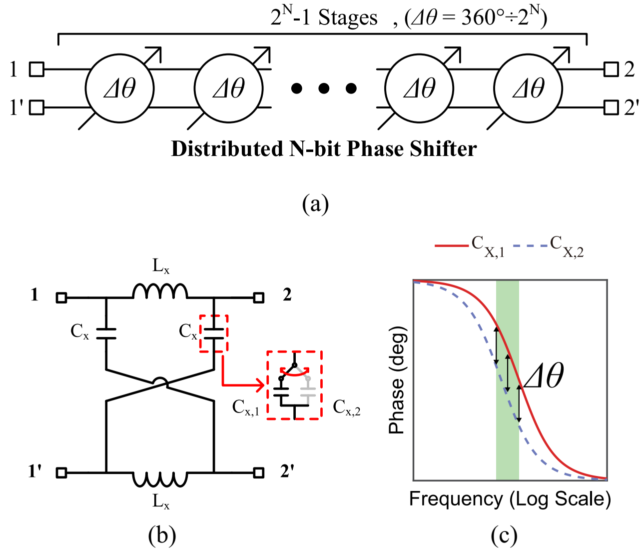

FIGURE 1. **(a) Configuration of a differential distributed N-bit phase shifter with cascaded unit phase shifters. (b) Unit phase shifter based on a lattice network. (c) Constant-phase shift region of the proposed lattice phase shifter.**

it in the *E/W*-band using the FD-SOI process. In addition, we theoretically analyzed the lattice phase shifter design to operate as a wideband constant-phase shifter rather than a delay line.

II. ANALYSIS OF THE LATTICE PHASE SHIFTER

A. UNIT LATTICE PHASE SHIFTER

Fig. 1(b) shows the digitally loaded unit differential phase shifter with a lattice network. This phase shifter creates a phase shift using a fixed inductor (L_X) and a switchable-capacitor (C_X) that alternates between $C_{X,1}$ and $C_{X,2}$. According to the image impedance theory [33], the characteristic impedance (Z_X) and the insertion phase (θ_X) of the lattice network can be expressed as follows:

$$Z_X = \sqrt{\frac{L_X}{C_X}} \tag{1}$$

$$\theta_X = 2\tan^{-1}\left(\omega\sqrt{L_X C_X}\right). \tag{2}$$

Given that the switchable-capacitor (C_X) can be switched between $C_{X,1}$ and $C_{X,2}$, the insertion phase difference is

$$\begin{aligned} \Delta\theta &= 2\tan^{-1}\left(\omega\sqrt{L_X C_{X,2}}\right) - 2\tan^{-1}\left(\omega\sqrt{L_X C_{X,1}}\right) \\ &= 2\tan^{-1}\left(\frac{\omega\left(\sqrt{L_X C_{X,2}} - \sqrt{L_X C_{X,1}}\right)}{1+\omega^2 L_X\sqrt{C_{X,1}C_{X,2}}}\right). \end{aligned} \tag{3}$$

The tangent function can be approximated by a linear function when the input domain is sufficiently small. In (3), $\Delta\theta/2$ represents the input domain of the tangent function. If $\Delta\theta/2 = \pi/16$, which is sufficiently small value, then (3) can be expressed as follows:

$$\Delta\theta \approx \frac{2\omega\left(\sqrt{L_X C_{X,2}} - \sqrt{L_X C_{X,1}}\right)}{1+\omega^2 L\sqrt{C_{X,1}C_{X,2}}}. \tag{4}$$

When C_X is switched between $C_{X,1}$ and $C_{X,2}$, the characteristic impedances of the lattice network are Z_1 and Z_2 according to (1). Although it is impossible for both Z_1 and Z_2 to match Z_o, it is possible to have the same reflection coefficient. This equivalence allows both states to have the same return loss. The equation for both Z_1 and Z_2 to have the same reflection coefficient is as follows:

$$Z_o = \sqrt{Z_1 Z_2}. \tag{5}$$

From (1), the values of $C_{X,1}$ and $C_{X,2}$ that satisfy (5) are defined as follows:

$$C_X = \sqrt{C_{X,1} C_{X,2}}. \tag{6}$$

To identify the frequency range in which the phase shifter exhibits wideband characteristics, (4) is differentiated as follows:

$$\begin{aligned}\frac{d\Delta\theta}{d\omega} &= 2\frac{\left(\sqrt{L_X C_{X,2}} - \sqrt{L_X C_{X,1}}\right)\left(1 - \omega^2 L_X \sqrt{C_{X,1} C_{X,2}}\right)}{\left(1 + \omega^2 L_X \sqrt{C_{X,1} C_{X,2}}\right)^2} \\ &= 2\frac{\left(\sqrt{L_X C_{X,2}} - \sqrt{L_X C_{X,1}}\right)\left(1 - \omega^2 L_X C_X\right)}{\left(1 + \omega^2 L_X C_X\right)^2}.\end{aligned} \tag{7}$$

According to (6), the solution of (7) occurs when

$$\omega = 1/\sqrt{L_X C_X} \tag{8}$$

$$\theta_X|_{\omega=1/\sqrt{L_X C_X}} = 2tan^{-1}(1) = 90^\circ. \tag{9}$$

Given that $Z_X = Z_o$ and $\theta_X = 90^\circ$, L_X and C_X are determined by (1) and (2). In addition, for 22.5° unit phase shifter, $C_{X,1}$ and $C_{X,2}$ are determined by (4) and (6). In summary, a lattice phase shifter maintains consistent impedance matching regardless of the phase shift state and exhibits a flat phase shift over a broad frequency band.

B. UNIT LADDER PHASE SHIFTER

Fig. 2(a) shows the digitally loaded unit differential phase shifter, which features a ladder network. This type of phase shifter creates a phase shift using a fixed inductor (L_π) and a switchable-capacitor (C_π) that can switch between $C_{\pi,1}$ and $C_{\pi,2}$. However, Fig. 2(b) shows that the ladder phase shifter exhibits inconsistent phase shift characteristics compared with the lattice phase shifter.

Using the ABCD matrix [17], [18], [19], the characteristic impedance (Z_π) and the insertion phase (θ_π) of the ladder network can be expressed as follows:

$$Z_\pi = \sqrt{\frac{L_\pi}{C_\pi\left(1 - \omega^2 L_\pi C_\pi\right)}} \tag{10}$$

$$\theta_\pi = 2\sin^{-1}\left(\omega\sqrt{L_\pi C_\pi}\right). \tag{11}$$

To determine the values of L_π and C_π, these equations can be expressed as follows:

$$L_\pi = \frac{Z_\pi \sin|\theta_\pi|}{2\omega} \tag{12}$$

$$C_\pi = \frac{\tan|\theta_\pi/2|}{\omega Z_\pi}. \tag{13}$$

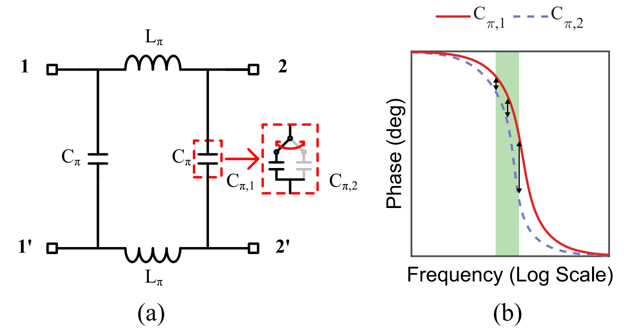

FIGURE 2. (a) Configuration of a unit phase shifter based on a ladder network. (b) Inconsistent phase shift region of the conventional ladder phase shifter.

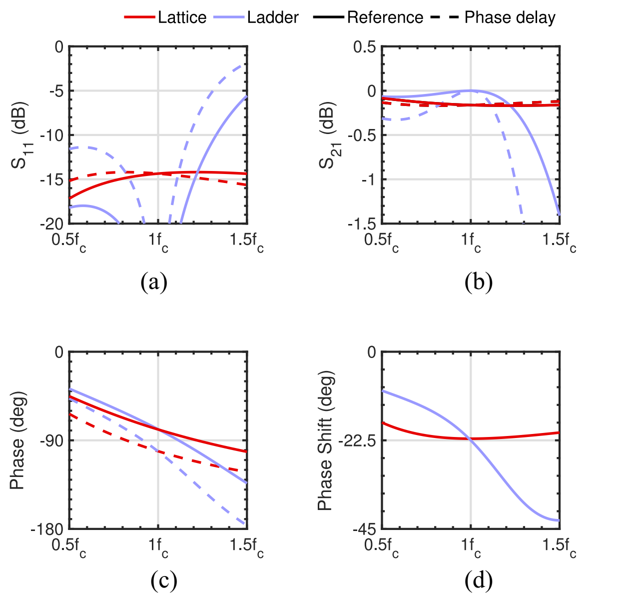

FIGURE 3. Simulated reference state (solid line) and phase delay state (dashed line) of ideal lattice- and ladder-network. (a) Return loss. (b) Insertion loss. (c) Phase. (d) Relative phase shift.

Given that $\sin|\theta_\pi|$ in (12) is symmetric with respect to $\theta_\pi = 90^\circ$, the value of L_π remains constant whether θ_π is $90^\circ + \Delta\theta/2$ or $90^\circ - \Delta\theta/2$. By applying (13), $C_{\pi,1}$ and $C_{\pi,2}$ can be determined when θ_π is $90^\circ + \Delta\theta/2$ and $90^\circ - \Delta\theta/2$, respectively. Notably, the characteristic impedance (Z_π) of the ladder network remains constant when θ_π is either $90^\circ + \Delta\theta/2$ or $90^\circ - \Delta\theta/2$. This consistency persists even if the value of the switchable-capacitor (C_π) changes, as shown in (10) and (13).

C. S-PARAMETER ANALYSIS: LATTICE VERSUS LADDER

In Fig. 3, the S-parameters of unit phase shifters using lattice and ladder networks are shown in the reference state ($C_{X,1}$, $C_{\pi,1}$) and phase delay state ($C_{X,2}$, $C_{\pi,2}$). At the central frequency, the ladder phase shifter has perfect impedance matching as shown in Fig. 3(a). However, (1) and (10) reveal that the lattice phase shifter exhibits superior frequency-independent impedance matching compared with the ladder

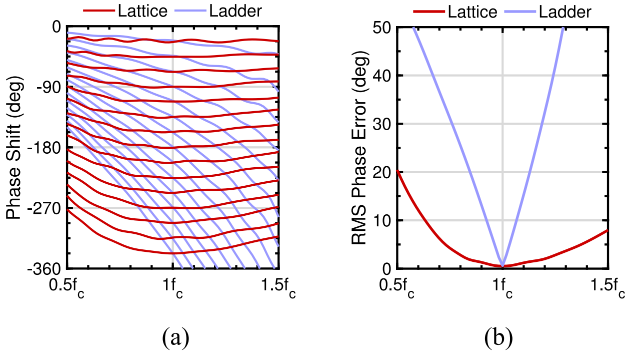

FIGURE 4. Simulated (a) phase shift and (b) RMS phase error of the distributed phase shifters using lattice- and ladder-network.

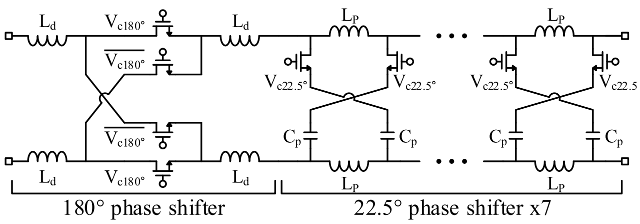

FIGURE 5. Configuration of the proposed digitally loaded differential phase shifter using lattice networks.

phase shifter. The difference between these two network structures can also be seen in insertion loss, as shown in Fig. 3(b). Although both lattice and ladder phase shifters share the same insertion phase according to states ($\theta = 90° \pm \Delta\theta/2$), Fig. 3(d) demonstrates that the lattice phase shifter maintains a flat relative phase shift over a wide bandwidth, whereas the phase shifter of the ladder phase shifter varies with frequency.

The simulation results in Fig. 4 show 360° phase shifters that are cascaded with a ladder network and a lattice network, respectively. Although both phase shifters can be designed with an RMS phase error close to 0° at the center frequency, only the lattice phase shifter shows low phase error within a wide frequency band. This is because the two unit phase shifters have different phase shift characteristics, as shown in Fig. 3(d). In summary, comparing the two distributed phase shifters shows that the lattice phase shifter is a more suitable option for wideband operation. In a real-world design, the parasitic shunt capacitance of the lattice structure causes the circuit to behave like a combination of an ideal lattice and ladder network. Therefore, we optimized the size of the lattice capacitors and inductors to offset the effects of the parasitic shunt capacitance.

III. IMPLEMENTATION

Fig. 5 shows a schematic of the proposed wideband phase shifter, which comprises a double-pole double-throw switch (DPDT) operating as a 180° phase shifter and seven lattice phase shifters operating as 22.5° phase shifters. The proposed passive phase shifter features a 4-bit phase resolution,

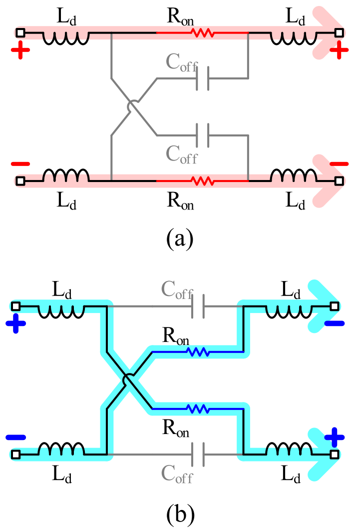

FIGURE 6. Configuration of the proposed 180° phase shifter in (a) in-phase state and (b) out-of-phase state.

which is a reasonable choice when considering both phase resolution and insertion loss. The 4-bit phase shifter, which is the most widely studied phase shifter, exhibits < 0.1-dB loss in the scanning direction (16-element phased array system) [16]. Furthermore, implementing phase steps below 22.5° (11.25°, 5.625°, etc.) is relatively easy with a wideband phase shifter. In addition, phase shifters with small phase steps can be designed as continuous phase shifter using varactors. Therefore, distributed phase shifters can easily achieve high phase resolution with additional cascading.

The DPDT switch offers an optimal structure for a 180° phase shifter due to its low loss and constant-phase shift across the entire bandwidth, making it suitable for differential phase shifter applications. The 180° phase shifter employs differential CMOS quad switches to create the in-phase and out-of-phase states by switching the signal lines ($V_{c180°}$).

Fig. 6 shows the in-phase and out-of-phase states of the 180° phase shifter. Input and output inductors were used to resonate out the off-capacitance (C_{OFF}) of the transistors. The differential signal of a 180° phase shifter is guided to in-phase [Fig. 6(a)] or out-of-phase [Fig. 6(b)] through the small on-resistance (R_{ON}) of the transistor.

In the 22.5° phase shifters, switchable-capacitors introduce the reference and phase delay states by changing the capacitance of the lattice network. Fig. 7 shows a schematic of the proposed lattice phase shifter based on transistor operation. When the transistor is switched off, the transistor's off-capacitance is in series with C_p, resulting in a reduced capacitance as shown in Fig. 7(a). When the transistor is switched on, the transistor's on-resistance is in series with the capacitor C_p as shown in Fig. 7(b). By changing the capacitance of the lattice phase shifter through transistor operation, a 22.5° phase shifting is achieved. To minimize the change in insertion loss due to phase shifter operation, the lowest on-resistance of the available transistor models was used for each process.

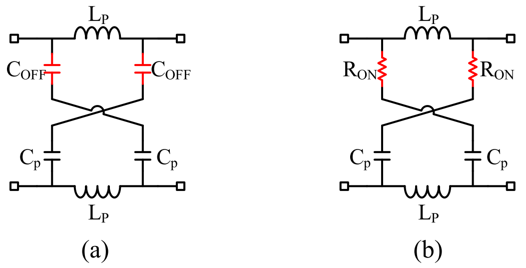

FIGURE 7. Configuration of the proposed lattice phase shifter in (a) reference state and (b) phase delay state.

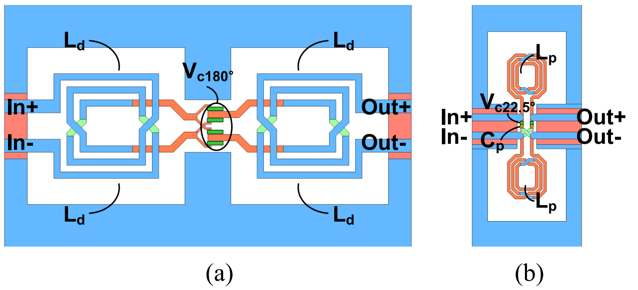

FIGURE 8. EM-simulation layout of *K/Ka*-band. (a) 180° DPDT phase shifter. (b) 22.5° lattice phase shifter.

A. K/KA-BAND PHASE SHIFTER

The *K/Ka*-band phase shifter used low-threshold triple-well transistors in a 28-nm bulk CMOS process. The DPDT switch requires two coupled inductors for differential operation. As shown in Fig. 8(a), these coupled inductors (L_d = 187 pH, K = 0.4) are used to reduce the size of the DPDT switch and used as a matching inductor. The width of the nMOS switches controlled by $V_{c180°}$ was 20 um. The EM-simulation layout of the lattice phase shifter is shown in Fig. 8(b). In the 22.5° lattice phase shifter, the inductor L_p and capacitor C_p were 580 pH and 47 fF, respectively. The width of the nMOS switches controlled by $V_{c22.5°}$ was 50 um. The input-referenced 1-dB compression point was 5 dBm in the simulations.

B. E/W-BAND PHASE SHIFTER

The *E/W*-band phase shifter used extended-gate transistors in a 28-nm FD-SOI CMOS process. A DPDT switch requires four matching inductors. In the *E/W*-band, the parasitic capacitance of the inductors had to be minimized. Therefore, four inductors without mutual coupling were employed (L_d = 104 pH), as shown in Fig. 9(a). The width of the nMOS switches controlled by $V_{c180°}$ was 23 um. Fig. 9(b) shows the EM-simulation layout of the lattice phase shifter. In the 22.5° lattice phase shifter, the inductor L_p and capacitor C_p were 227 pH and 26.5 fF, respectively. The width of the nMOS switches controlled by $V_{c22.5°}$ was 50 um. The input-referenced 1-dB compression point was 9.2 dBm in the simulations, which is relatively higher than the 5-dBm input-referenced 1-dB compression point of the *K/Ka*-band

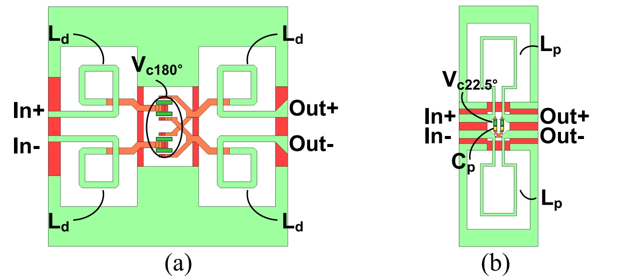

FIGURE 9. EM-simulation layout of *E/W*-band. (a) 180° DPDT phase shifter. (b) 22.5° lattice phase shifter.

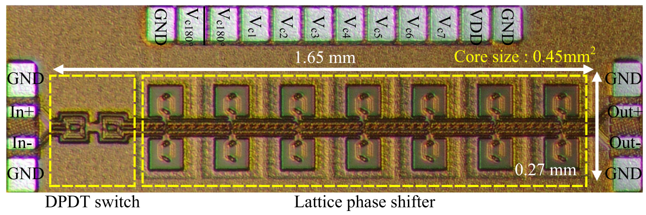

FIGURE 10. Chip microphotograph of the *K/Ka*-band digitally loaded differential phase shifter using lattice networks.

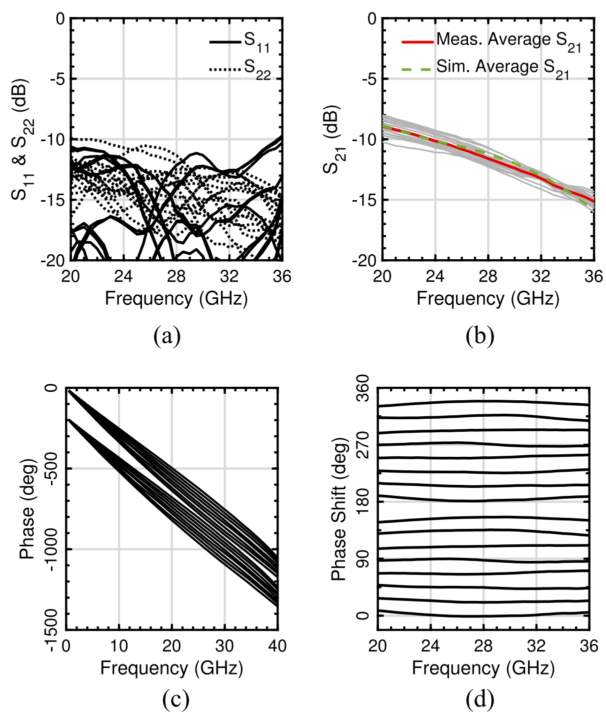

FIGURE 11. Measured (a) input and output return losses, (b) insertion loss, (c) insertion phase, and (d) relative phase shift of 16 different phase states of the *K/Ka*-band phase shifter.

phase shifter. This result indicates that FD-SOI transistors have superior linear performance compared to bulk CMOS.

IV. MEASUREMENTS

A. K/KA-BAND PHASE SHIFTER

The chip photograph of the *K/Ka*-band phase shifter with a size of 1.65 mm × 0.27 mm (0.45 mm^2) without pads is

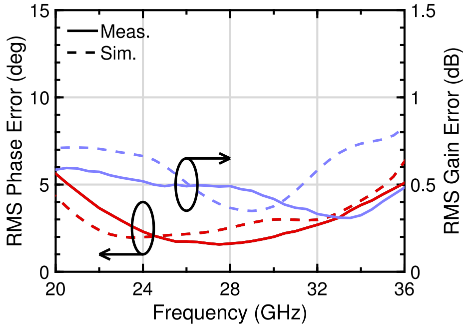

FIGURE 12. Measured RMS phase and gain errors of the *K/Ka*-band phase shifter.

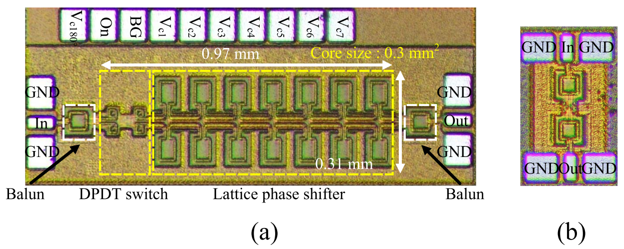

FIGURE 13. Chip microphotograph of the (a) *E/W*-band digitally loaded differential phase shifter using lattice networks and (b) back-to-back single-to-differential balun.

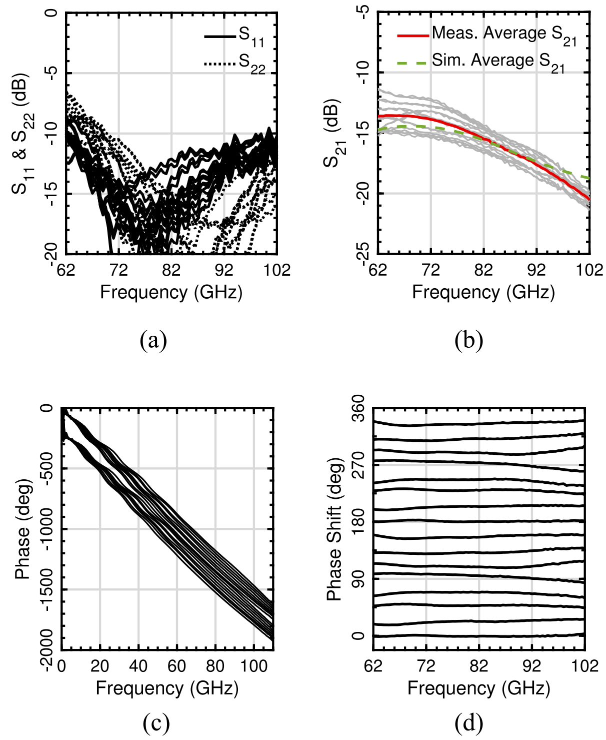

FIGURE 14. Measured (a) input and output return losses, (b) insertion loss, (c) insertion phase, and (d) relative phase shift of 16 different phase states of the *E/W*-band phase shifter.

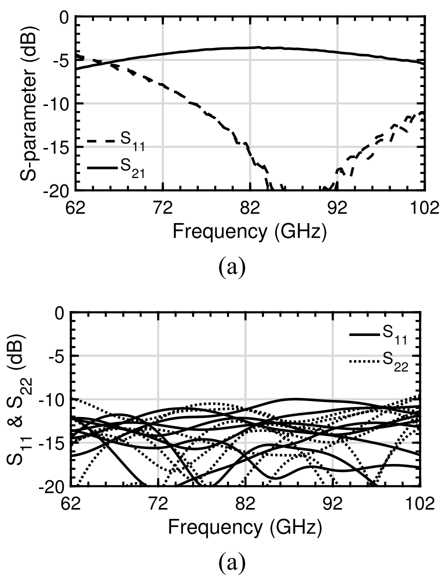

FIGURE 15. (a) Measured S-parameters of the back-to-back *E/W*-band balun. (b) Simulated input and output return losses of 16 different phase states of the *E/W*-band phase shifter.

shown in Fig. 10. A 4-port VNA and GSSG probes were used for measurements in the *K/Ka*-band. Fig. 11(a) shows the measured input and output return losses for all 16 states. Both input and output return losses are larger than 10 dB from 20.5 to 35.5 GHz. Fig. 11(b) shows the insertion losses for all 16 states, with an average insertion loss of 11.6 ± 0.8 dB at 28 GHz. Fig. 11(c) shows the measured insertion phases of the 16 different phase states, clearly illustrating the phase characteristics of the lattice phase shifter and DPDT switches. Fig. 11(d) shows the measured relative phase shifts for all 16 states. The maximum error of the measured relative phase shift was −9.2 to +8.9° within 20.5–35.5 GHz. Fig. 12 shows the phase and gain errors calculated based on the standard deviation. The RMS phase error is from 1.6 to 5° within 20.5–35.5 GHz. The RMS gain error is from 0.3 to 0.6 dB within 20.5–35.5 GHz. These measurement error values demonstrate the capability of the proposed phase shifter to maintain constant-phase shift performance over a wide frequency band.

B. E/W-BAND PHASE SHIFTER

The chip photograph of the *E/W*-band phase shifter with a size of 0.97 mm × 0.31 mm (0.3 mm^2) without pads is shown in Fig. 13(a). A two-port VNA and GSG probes were employed for measurements in the *E/W*-band. To use the single-ended GSG probes, both ends of the circuit employed single-to-differential baluns. The performance of the baluns was verified using a back-to-back balun circuit that was separately implemented and measured under the same conditions as the *E/W*-band phase shifter, as shown in Fig. 13(b). Fig. 14(a) shows the measured input and output return losses for all 16 states, and both are larger than 10 dB from 69.5 to 101.5 GHz. Fig. 14(b) shows the insertion losses for all 16 states. The average insertion loss is 15.5 dB, with a loss variation of ±1.1 dB at 82 GHz. It is important to note that these results include the balun

TABLE 1. Performance comparison of reported 360° phase shifters.

Reference	**This Work**		TCASI 2022 [14]	TCASI 2021 [13]	TCASI 2023 [15]	TCASII 2020 [9]	MWCL 2021 [12]	MWCL 2020 [17]	JSSC 2008 [16]	IMS 2018 [25]
Process	28-nm CMOS	28-nm FD-SOI	65-nm CMOS	40-nm CMOS	65-nm CMOS	0.13-μm SiGe BiCMOS	130-nm CMOS	28-nm CMOS	0.12-μm SiGe BiCMOS	0.12-μm SiGe BiCMOS
Type	Lattice PS	Lattice PS	P-VMPS	P-VMPS	P-VMPS	A-VMPS	P-VMPS	STPS	STPS	RTPS + PI$^{\oplus}$
Bi-direction	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	No
FBW (%) /(GHz)	**53.6** /20.5∼35.5	**45.1** /63.5∼100.5	**27** /32∼42	**19.4** /72∼87.5	**14.3** /130∼150	**13.7** /75∼86	**37.5** /26∼38	**8.2** /35∼38	**14.3** /32.5∼37.5	**9** /117*∼128
RMS phase err. (deg)	1.6∼5	2.4∼4.6	0.45∼2.8	1∼2.4*	1.1∼5.5*	1.35∼6‡	1.5*∼3	1.2∼8*	4∼**10***	2.2∼**10***
RMS gain err. (dB)	0.3∼0.6	0.44∼**1**	0.2∼0.45*	0.3∼**1***	0.6∼**1***	0.46∼**1***	0.4*∼0.6	0.9*∼**1***	0.7*∼**1***	0.1*∼0.4*
Return loss (dB)	**10**	**10**$^{\diamond}$	**10**	11	11*	**10**	**10**	**10**	11*	12*
Phase step (deg)	22.5	22.5	2.8125	5.625	5.625	5.625	5.625	22.5	22.5	11.25
Average gain @f_0 (dB)	-11.6±0.8	-11.9±1.1$^{\diamond}$	-17±0.4	-15.1±0.3*	-16.5±2.3	7±1.1*	-10±0.5	-12.8±2.5	-13±1.1	-5.8±0.6
DC power (mW)	0	0	0	0	0	60	0	0	0	30
Size (mm^2)	0.45	0.3$^{\diamond}$	0.14	0.15	0.03	0.31	0.15	0.08	0.12	0.41

FBW criteria : RMS phase error $<$ 10°, RMS gain error $<$ 1 dB, RL $>$ 10 dB.
Parameters determining the FBW have been highlighted in bold.
$^{\diamond}$Without baluns. *Estimated from figures. $^{\oplus}$ Phase inverter ‡ Results over 70 $\sim$ 85 GHz.

used for measurement. Fig. 15(a) shows the measured input and output return losses of the back-to-back balun larger than 10 dB across the frequency range of 75–105 GHz. In addition, the insertion loss of the back-to-back balun is 3.6 dB at 82 GHz. Considering this insertion loss, the average insertion loss of the proposed differential phase shifter is 11.9 dB at 82 GHz. In Fig. 15(b), the simulated input and output return losses of the proposed phase shifter are shown without the incorporation of baluns. Both input and output return losses are larger than 10 dB from 63.5 to 100.5 GHz. Therefore, the baluns limit the range of frequencies where the measured return loss is larger than 10 dB and mask the expected multiple resonance behavior of the distributed structure. Fig. 14(c) shows the measured insertion phases of 16 different phase states. Fig. 14(d) shows the measured relative phase shifters of all 16 states. The maximum error of the measured relative phase shift was -10.0 to +8.8° within 63.5–100.5 GHz. Fig. 16 shows the phase and gain errors calculated based on the standard deviation. The RMS phase error ranges from 2.4 to 4.6° within 63.5–100.5 GHz. The RMS gain error ranges from 0.44 to 1 dB within 63.5–100.5 GHz. Similar to the *K/Ka*-band measurements, the RMS phase error and gain error demonstrate that the proposed phase

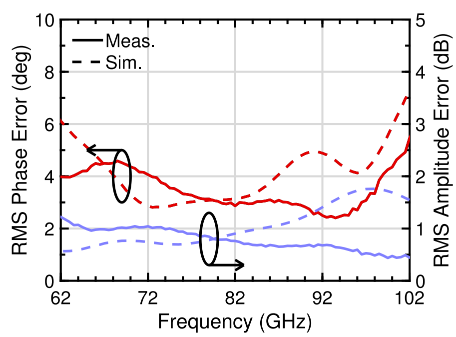

FIGURE 16. Measured RMS phase and gain errors of the *E/W*-band phase shifter.

shifter exhibits wideband characteristics regardless of the process and operating band.

C. BANDWIDTH COMPARISON

Table 1 compares the proposed phase shifters with various 360° phase shifters. Previously, comparing the fractional bandwidth (FBW) of different phase shifters has been challenging because of their varying operating frequency bands chosen based on their individual design criteria. To facilitate the comparison of various phase shifters, a common bandwidth criterion was defined. The key specifications for a constant-phase shifter include the RMS phase/gain error and return loss, which directly affect the accurate beamforming of multistandard phased array systems. Consequently, the FBW of the phase shifter was determined to ensure an RMS phase error of less than 10°, an RMS gain error of less than 1 dB, and a return loss of more than 10 dB. This common FBW criterion enables the comparison of phase shifters of different specifications. The RMS phase/gain error and return loss values presented in Table 1 are the values within the FBW. In addition, the parameters determining the FBW have been highlighted in bold.

The proposed *K/Ka*- and *E/W*-band phase shifters achieve significantly wider FBWs of 53.6% and 45.1%, respectively, surpassing all existing phase shifters listed in Table 1. Although the proposed circuit exhibits a relatively large chip size, the wideband operation of the proposed circuit fulfills multiple standards, reducing the need to design individual circuit blocks based on frequency and lowering the overall cost of deployment. In addition, the design enables bidirectional operation, offering potential size reduction in phased array systems by sharing a single bidirectional phase shifter between TX and RX [14], [34]. Although passive phase shifters typically exhibit higher insertion loss than active phase shifters, this can be effectively compensated by the implementation of a single amplifier. This approach offers improved efficiency in terms of power consumption and superior linear output compared with A-VMPS, which often employs multiple VGAs and suffers from an inherent loss of 3 dB in the combiner [13].

V. CONCLUSION

In this article, we introduce an ultra-wideband 4-bit distributed lattice phase shifter. To achieve a wider bandwidth than conventional designs, the distributed phase shifter uses an all-pass lattice network instead of the typical low-pass CLC-π ladder network. Using a DPDT switch and seven lattice phase shifters, the proposed phase shifter achieves a full 360° phase shift range. The *K/Ka*-band phase shifter achieved an FBW of 53.6% with an RMS phase error of less than 5° and a low RMS gain error of less than 0.6 dB. Similarly, the *E/W*-band phase shifter achieved an FBW of 45.1% with RMS phase error less than 4.6° and RMS gain error less than 1 dB. The proposed phase shifters exhibited the widest FBW among Table 1. In addition, the low RMS error within this FBW demonstrates the proposed phase shifter for high-accuracy multistandard operations.

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REFERENCES

[1] M. Oveisi, H. Wang, and P. Heydari, "A study of a millimeter-wave transmitter architecture realizing QAM directly in RF domain," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 70, no. 6, pp. 2243–2256, Jun. 2023, doi: 10.1109/TCSI.2023.3255109.

[2] O. Memioglu, Y. Zhao, and B. Razavi, "A 300GHz 52mW CMOS receiver with on-chip LO generation," in *Proc. IEEE Int. Solid-State Circuits Conf. (ISSCC)*, vol. 65, Feb. 2022, pp. 1–3, doi: 10.1109/ISSCC42614.2022.9731745.

[3] P. Heydari, "Terahertz integrated circuits and systems for high-speed wireless communications: Challenges and design perspectives," *IEEE Open J. Solid-State Circuits Soc.*, vol. 1, pp. 18–36, 2021, doi: 10.1109/OJSSCS.2021.3110748.

[4] D. Zhao, J. Zhang, Y. Yi, P. Gu, and N. Jiang, "5G millimeter-wave phased-array transceiver: System considerations and circuit implementations," in *Proc. IEEE Int. Symp. Circuits Syst. (ISCAS)*, May 2019, pp. 1–4, doi: 10.1109/ISCAS.2019.8702530.

[5] N. Imai and H. Ichikawa, "One-chip endless phase shifter IC's for space diversity combiner," *IEEE Trans. Circuits Syst. II, Analog Digit. Signal Process.*, vol. 43, no. 4, pp. 281–288, Apr. 1996, doi: 10.1109/82.488282.

[6] K.-J. Koh and G. M. Rebeiz, "0.13-μm CMOS phase shifters for X-, Ku-, and K-band phased arrays," *IEEE J. Solid-State Circuits*, vol. 42, no. 11, pp. 2535–2546, Nov. 2007, doi: 10.1109/JSSC.2007.907225.

[7] D. Pepe and D. Zito, "Two mm-Wave vector modulator active phase shifters with novel IQ generator in 28 nm FDSOI CMOS," *IEEE J. Solid-State Circuits*, vol. 52, no. 2, pp. 344–356, Feb. 2017, doi: 10.1109/JSSC.2016.2605659.

[8] S. Afroz and K.-J. Koh, "90° hybrid-coupler based phase-interpolation phase-shifter for phased-array applications at W-band and beyond," in *Proc. IEEE MTT-S Int. Microw. Symp. (IMS)*, May 2016, pp. 1–4, doi: 10.1109/MWSYM.2016.7540415.

[9] H. Li, J. Chen, D. Hou, and W. Hong, "A W-Band 6-bit phase shifter with 7 dB gain and 1.35° RMS phase error in 130 nm SiGe BiCMOS," *IEEE Trans. Circuits Syst. II, Exp. Briefs*, vol. 67, no. 10, pp. 1839–1843, Oct. 2020, doi: 10.1109/TCSII.2019.2944166.

[10] G. H. Park, C. W. Byeon, and C. S. Park, "A 60-GHz low-power active phase shifter with impedance-invariant vector modulation in 65-nm CMOS," *IEEE Trans. Microw. Theory Techn.*, vol. 68, no. 12, pp. 5395–5407, Dec. 2020, doi: 10.1109/TMTT.2020.3023705.

[11] W. Gao and D. Zhao, "K-band 360° passive vector modulator phase shifter with coupled line qadrature coupler and passive transistor array," in *Proc. IEEE MTT-S Int. Wireless Symp. (IWS)*, May 2021, pp. 1–3, doi: 10.1109/IWS52775.2021.9499549.

[12] S. Londhe and E. Socher, "28–38-GHz 6-bit compact passive phase shifter in 130-nm CMOS," *IEEE Microw. Wireless Compon. Lett.*, vol. 31, no. 12, pp. 1311–1314, Dec 2021, doi: 10.1109/LMWC.2021.3112689.

[13] P. Gu, D. Zhao, and X. You, "Analysis and design of a CMOS bidirectional passive vector-modulated phase shifter," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 68, no. 4, pp. 1398–1408, Apr. 2021, doi: 10.1109/TCSI.2020.3048816.

[14] Y. Li et al., "A 32–40 GHz 7-bit bi-directional phase shifter with 0.36 dB/1.6° RMS magnitude/phase errors for phased array systems," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 69, no. 10, pp. 4000–4013, Oct 2022, doi: 10.1109/TCSI.2022.3188157.

[15] Z. Zhang, M. Yang, X. Geng, K. Wang, Q. Xie, and Z. Wang, "A 0.0325-mm^2 114-to-147-GHz 6-bit passive vector-modulated phase shifter with MN-embedded isolated power combiner achieving $<$ 3.7° RMS phase error in 65-nm CMOS," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 70, no. 11, pp. 4394–4405, Nov. 2023, doi: 10.1109/TCSI.2023.3300033.

[16] B.-W. Min and G. M. Rebeiz, "Single-ended and differential Ka-band BiCMOS phased array front-ends," *IEEE J. Solid-State Circuits*, vol. 43, no. 10, pp. 2239–2250, Oct. 2008, doi: 10.1109/JSSC.2008.2004336.

[17] M. Jung and B.-W. Min, "A compact Ka-band 4-bit phase shifter with low group delay deviation," *IEEE Microw. Wireless Compon. Lett.*, vol. 30, no. 4, pp. 414–416, Apr. 2020, doi: 10.1109/LMWC.2020.2975108.

[18] T. Hancock and G. Rebeiz, "A 12-GHz SiGe phase shifter with integrated LNA," *IEEE Trans. Microw. Theory Techn.*, vol. 53, no. 3, pp. 977–983, Mar. 2005, doi: 10.1109/TMTT.2004.842479.

[19] C. F. Campbell and S. A. Brown, "A compact 5-bit phase-shifter MMIC for K-band satellite communication systems," *IEEE Trans. Microw. Theory Techn.*, vol. 48, no. 12, pp. 2652–2656, Dec. 2000, doi: 10.1109/22.899026.

[20] H.-S. Lee and B.-W. Min, "W-band CMOS 4-bit phase shifter for high power and phase compression points," *IEEE Trans. Circuits Syst. II, Exp. Briefs*, vol. 62, no. 1, pp. 1–5, Jan. 2015, doi: 10.1109/TCSII.2014.2362732.

[21] G.-S. Shin et al., "Low insertion loss, compact 4-bit phase shifter in 65 nm CMOS for 5G applications," *IEEE Microw. Wireless Compon. Lett.*, vol. 26, no. 1, pp. 37–39, Jan. 2016, doi: 10.1109/LMWC.2015.2505624.

[22] M. Elkholy, S. Shakib, J. Dunworth, V. Aparin, and K. Entesari, "Low-loss highly linear integrated passive phase shifters for 5G front ends on bulk CMOS," *IEEE Trans. Microw. Theory Techn.*, vol. 66, no. 10, pp. 4563–4575, Oct 2018, doi: 10.1109/TMTT.2018.2859994.

[23] T.-W. Li and H. Wang, "A millimeter-wave fully integrated passive reflection-type phase shifter with transformer-based multi-resonance loads for 360° phase shifting," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 65, no. 4, pp. 1406–1419, Apr. 2018, doi: 10.1109/TCSI.2017.2748078.

[24] P. Gu and D. Zhao, "Ka-band CMOS 360° reflective-type phase shifter with ±0.2 dB insertion loss variation using triple-resonating load and dual-voltage control techniques," in *Proc. IEEE Radio Freq. Integr. Circuits Symp. (RFIC)*, Jun. 2018, pp. 140–143, doi: 10.1109/RFIC.2018.8428987.

[25] R. B. Yishay and D. Elad, "D-band 360° phase shifter with uniform insertion loss," in *IEEE/MTT-S Int. Microw. Symp. Dig. Paper*, Jun. 2018, pp. 868–870, doi: 10.1109/MWSYM.2018.8439838.

[26] F. Ellinger, H. Jackel, and W. Bachtold, "Varactor-loaded transmission-line phase shifter at C-band using lumped elements," *IEEE Trans. Microw. Theory Techn.*, vol. 51, no. 4, pp. 1135–1140, Apr. 2003, doi: 10.1109/TMTT.2003.809670.

[27] Y. Tousi and A. Valdes-Garcia, "A Ka-band digitally-controlled phase shifter with sub-degree phase precision," in *Proc. IEEE Radio Freq. Integr. Circuits Symp. (RFIC)*, May 2016, pp. 356–359, doi: 10.1109/RFIC.2016.7508326.

[28] T.-S. Chu, J. Roderick, and H. Hashemi, "An integrated ultra-wideband timed array receiver in 0.13 μm CMOS using a path-sharing true time delay architecture," *IEEE J. Solid-State Circuits*, vol. 42, no. 12, pp. 2834–2850, Dec 2007, doi: 10.1109/JSSC.2007.908746.

[29] I. Mondal and N. Krishnapura, "A 2-GHz bandwidth, 0.25–1.7 ns true-time-delay element using a variable-order all-pass filter architecture in 0.13 μ m CMOS," *IEEE J. Solid-State Circuits*, vol. 52, no. 8, pp. 2180–2193, Aug. 2017, doi: 10.1109/JSSC.2017.2693229.

[30] S. Park and S. Jeon, "A 15–40 GHz CMOS true-time delay circuit for UWB multi-antenna systems," *IEEE Microw. Wireless Compon. Lett.*, vol. 23, no. 3, pp. 149–151, Mar. 2013, doi: 10.1109/LMWC.2013.2244872.

[31] S. Barker and G. Rebeiz, "Distributed MEMS true-time delay phase shifters and wide-band switches," *IEEE Trans. Microw. Theory Techn.*, vol. 46, no. 11, pp. 1881–1890, Nov. 1998, doi: 10.1109/22.734503.

[32] S. Kwon, M. Jung, and B.-W. Min, "Wideband switchable-capacitor loaded differential phase shifter with lattice structures," in *Proc. IEEE/MTT-S Int. Microw. Symp.*, Jun. 2022, pp. 738–741, doi: 10.1109/IMS37962.2022.9865273.

[33] P. Peres, C. Souza, and I. Bonatti, "ABCD matrix: A unique tool for linear two-wire transmission line modelling," *Int. J. Electr. Eng. Educ.*, vol. 40, no. 3, p. 5, 2003, doi: 10.7227/IJEEE.40.3.5.

[34] U. Kodak and G. M. Rebeiz, "A 5G 28-GHz common-Leg T/R front-end in 45-nm CMOS SOI with 3.7-dB NF and −30-dBc EVM with 64-QAM/500-MBaud modulation," *IEEE Trans. Microw. Theory Techn.*, vol. 67, no. 1, pp. 318–331, Jan. 2019, doi: 10.1109/TMTT.2018.2873374.

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